

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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[Website](#) |

[Linkedin](#) |

[Google Scholar](#)

Education

Embry-Riddle Aeronautical University
M.S. in Aviation
Specialization in Aviation Safety Systems

Daytona Beach, FL
Aug 2025–Present
GPA 4.0/4.0

Embry-Riddle Aeronautical University
B.S. in Aeronautical Science
Minor in Air Transportation

Daytona Beach, FL
May 2023
GPA 3.85/4.0

Research Interests

Flight Safety Analytics; ADS-B Analysis; Approach Stability Assessment; Terminal Airspace Operations

Research Experience

Graduate Researcher

School of Graduate Studies, College of Aviation

Advisor: Dr. João Souza Dias Garcia

Sep 2025–Present

- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases

Graduate Researcher

Applied Aerospace Research & Development Lab

Advisor: Dr. Ryan Wallace

Feb 2026–Present

- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[1] Yoo, T. & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. (Accepted). American Institute of Aeronautics and Astronautics AVIATION Forum.

Presentations

- Characterizing Air Traffic Density Using Nationwide ADS-B Data. (Poster). Embry-Riddle Aeronautical University Discovery Day, 2026.
- Data-Driven Framework for Assessing Wind Impacts on Approach Stability in Flight Training. (Poster). BCAAS Third Annual Safety Research Symposium, 2026
- Approach Stability Assessment in Flight Training Using ADS-B Data. (Poster). Embry-Riddle Aeronautical University Student Research Symposium, 2025.

Teaching Experience

Graduate Teaching Assistant

Supervisor: Dr. Dahai Liu

Embry-Riddle Aeronautical University

Dec 2025–Present

Teaching support for:

- Statistical Analysis for Aviation/Aerospace
- Research Methods in Aviation/Aerospace
- Graduate Capstone Research Project

Flight Instructor

Instructional areas include:

- Private Pilot Certificate
- Instrument Rating
- Commercial Pilot Certificate
- Certified Flight Instructor
- Certified Flight Instructor-Instrument
- Flight Review and Instrument Proficiency Check
- Complex Endorsement

Aviators Academy
Nov 2023–July 2025

Honors and Awards

Best Poster Award, Embry-Riddle Aeronautical University

April 2026

1st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University

November 2025

Magna Cum Laude, Embry-Riddle Aeronautical University

May 2023

Deans List (6 semesters), Embry-Riddle Aeronautical University

Certifications and Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane

Flight Instructor – ASEL, AMEL, Instrument Airplane

Ground Instructor – Advanced, Instrument

Remote Pilot – sUAS

Programming Skills

Python (pandas, traffic, scikit-learn, Matplotlib)

SPSS (PROCESS macro)

Professional Memberships

American Institute of Aeronautics and Astronautics

University Aviation Association

Royal Aeronautical Society

Leadership & Service

Secretary

Korean Student Association

Embry-Riddle Aeronautical University
2019–2020

Administrative Specialist; Squad Commander

Republic of Korea Military Service

Korea Defense Communication Command
2017–2019

Vice Captain

Table Tennis Club

University of North Dakota
2016–2017